A Hybrid Computing-in-Memory Architecture for Energy-Efficient Edge AI Inference

Fangchao Lou[a], Dandan Chen[b], Jian Cao[a], Xing Zhang[a], Bo Zhang[a]
[a]Peking University, Beijing, China
[b]Anhui University, Hefei, China
Corresponding Author Email: caojian@ss.pku.edu.cn zhx@pku.edu.cn

2025 10th International Conference on Integrated Circuits and Microsystems (ICICM) | 979-8-3315-8849-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICICM66614.2025.11315955

***Abstract*—To address the "memory wall" bottleneck in traditional von Neumann architectures and meet the growing demand for energy-efficient edge AI inference, this paper proposes an SRAM-based hybrid Computing-in-Memory (CIM) architecture that integrates analog and digital computing units. By combining the high energy efficiency of Analog CIM (ACIM) in low-precision parallel operations (such as convolution) with the high accuracy of Digital CIM (DCIM) in precision-critical tasks (such as activation functions), this architecture achieves a balanced trade-off between efficiency and accuracy.**

The core innovations of this architecture lie in three aspects: hardware-aware design, optimized dataflow scheduling, and integration of security mechanisms. At the hardware level, it is customized for edge AI operators like convolution, fully leveraging SRAM's CMOS compatibility and parallel computing capabilities. For dataflow scheduling, it adopts sliding window-aware input caching and a "compute-while-update" pipelining mechanism to significantly reduce data movement overhead. Additionally, it incorporates Advanced Encryption Standard (AES) operations into the CIM framework to ensure the security of data processing.

Experimental validation based on YOLO model inference shows that compared with traditional designs, this architecture reduces data movement energy consumption by 37% and computation latency by 29%, providing a robust low-power solution for edge AI applications that require both high energy efficiency and secure operation.

***Index Terms*—CIM,SRAM-based hybrid,Edge AI inference**

I. Introduction

To overcome the "memory wall" bottleneck in conventional von Neumann architectures, Computing-in-Memory (CIM) has emerged as a promising paradigm. By integrating computation directly into memory arrays, CIM enables data-centric processing that significantly reduces data movement, improves system efficiency, and lowers energy consumption.

As shown in Fig. 1, SRAM-based CIM extends traditional 6T cells to 8T or 10T structures, enabling in-situ multiply-and-accumulate (MAC) operations [1], [2]. These operations are executed in the analog domain and digitized via on-chip ADCs, allowing massively parallel matrix-vector multiplication within the memory array—ideal for accelerating AI workloads.

Unlike traditional accelerators, CIM systems broadcast input vectors while keeping weights stationary, minimizing data transfers and reducing latency and power. As illustrated in Fig. 2, this makes CIM highly suitable for edge AI inference

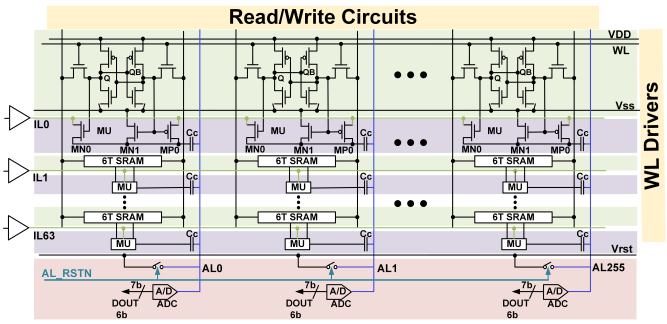


Fig. 1. Schematic Diagram of SRAM CIM Architecture

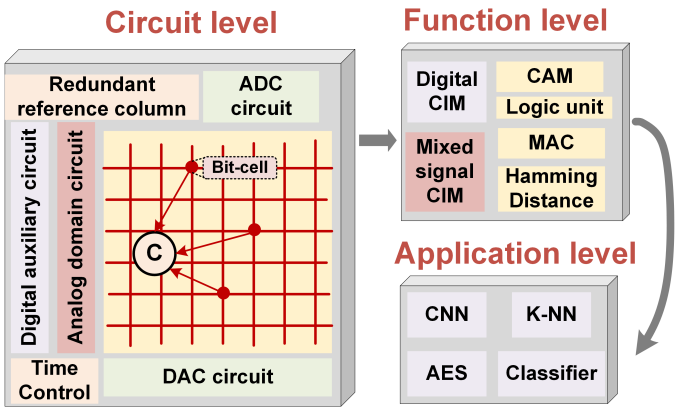


Fig. 2. Architectural Overview of CIM

tasks such as convolutional neural networks (CNN) [3], [4], where real-time and low-power processing is essential.

Recent CIM research includes emerging memory technologies (e.g., ReRAM, PCM, MRAM), analog/mixed-signal circuits, and architecture-level AI optimization [5]–[7]. Among them, SRAM-CIM stands out for its CMOS compatibility and speed, making it well-suited for edge deployment.

This paper proposes a hybrid SRAM-CIM architecture that integrates analog and digital MAC units, optimized dataflow scheduling, and hardware-aware quantization to achieve a balance between accuracy and energy efficiency. It offers a robust, low-power inference solution for edge AI applications [8].

The main contributions are:

1)A high-efficiency SRAM-CIM architecture optimized for

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:04:38 UTC from IEEE Xplore. Restrictions apply.

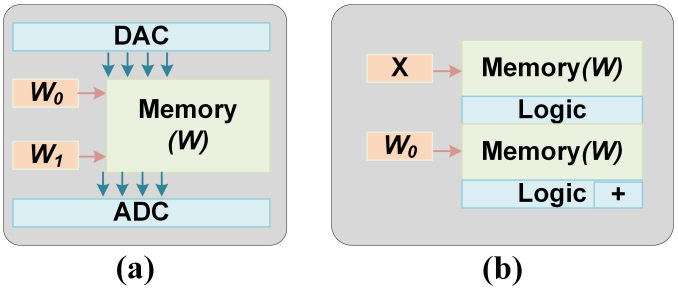


Fig. 3. (a)ACIM structure (b)DCIM structure

edge AI operators such as convolution and fully connected layers;

2)A hardware-aware quantization and dataflow reuse scheme tailored for CIM hardware constraints;

3)A hybrid analog-digital MAC design validated through YOLO-based inference, showing superior energy efficiency and robustness.

The rest of the paper is organized as follows: Section II introduces the proposed architecture. Section III presents experiments and case studies. Section IV concludes the paper.

II. Proposed Architecture

A. Overall Chip Architecture

The hybrid CIM architecture proposed in this study integrates the strengths of both Analog CIM (ACIM) and Digital CIM (DCIM) to achieve a balanced trade-off between energy efficiency and computational accuracy at both the circuit and system levels.

As shown in Fig. 3, from a technical perspective, ACIM performs MAC operations by leveraging the physical properties of memory media [9], demonstrating excellent energy efficiency in low-precision scenarios. It features a crossbar structure for high-parallelism computation, with simple hardware and low data movement energy. However, its performance is susceptible to noise, device variation, and limited precision. In contrast, DCIM relies entirely on digital logic using multipliers and adder trees to achieve high-precision results with strong noise immunity and controllable accuracy. Nonetheless, its hardware cost increases quadratically with precision [10], leading to higher power and area consumption.

As illustrated in Fig. 4, the proposed hybrid architecture utilizes ACIM within compute cores to handle low-precision, parallel operations such as convolutions, leveraging its high energy efficiency. DCIM modules are integrated across cores to correct the precision degradation of analog computations and support tasks that require higher accuracy [11]. The rationale for this hybrid design, as shown in Fig. 5, is based on the following observations:

When computational precision is below 8-bit, ACIM achieves significantly higher energy efficiency. Meanwhile, DCIM can digitally compensate for accuracy loss, supporting diverse precision requirements in edge AI tasks.

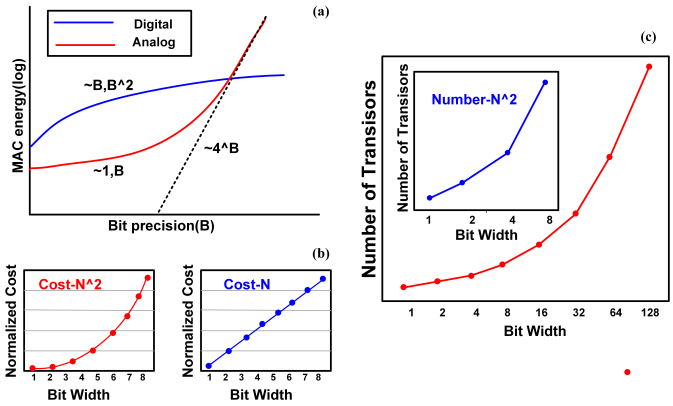


Fig. 4. (a)Trend of Overhead Variation in ACIM and DCIM
(b) Hardware overhead of bit-serial and bit-parallel multiplication in DCIM
(c)Accumulation Overhead in DCIM

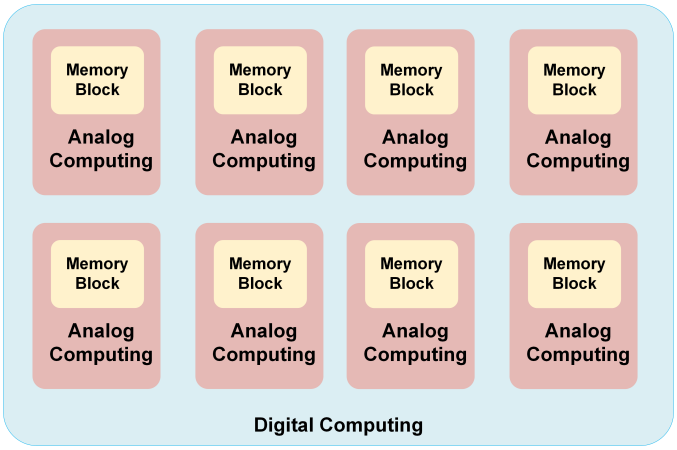


Fig. 5. Hybrid Digital-Analog Computing Architecture

The low area cost of ACIM, combined with the scalable nature of DCIM, enables the architecture to maximize computational density within limited chip area [12].

In CNN, where convolution operations account for over 50% of total workload, ACIM can directly perform matrix-vector multiplication within memory arrays, while DCIM is well-suited for high-precision operations such as activation functions and normalization—enabling efficient end-to-end acceleration.

This work proposes a high-efficiency edge AI chip architecture centered around an SRAM-based CIM accelerator, as depicted in Fig. 6. The overall structure follows the classical System-on-Chip (SoC) design paradigm, where a CPU serves as the core, interconnected via a central data bus to subsystems including memory, digital peripherals, and analog peripherals [13].

The key difference from traditional SoC lies in the inclusion of a CIM Accelerator based on SRAM in-memory computing. This accelerator features a data cache for weight storage, a CIM computation core, and an AHB bus controller for

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:04:38 UTC from IEEE Xplore. Restrictions apply.

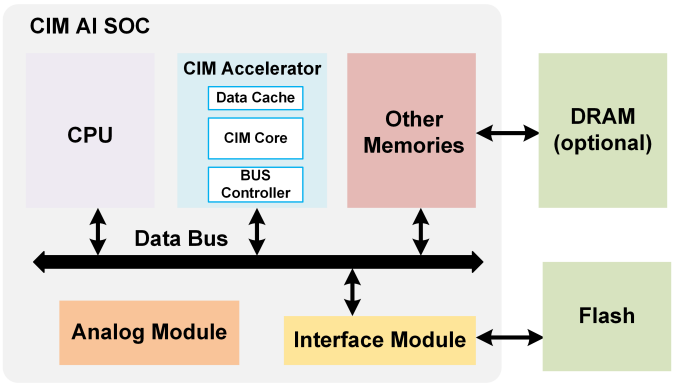


Fig. 6. CIM-Based Edge AI Chip Architecture

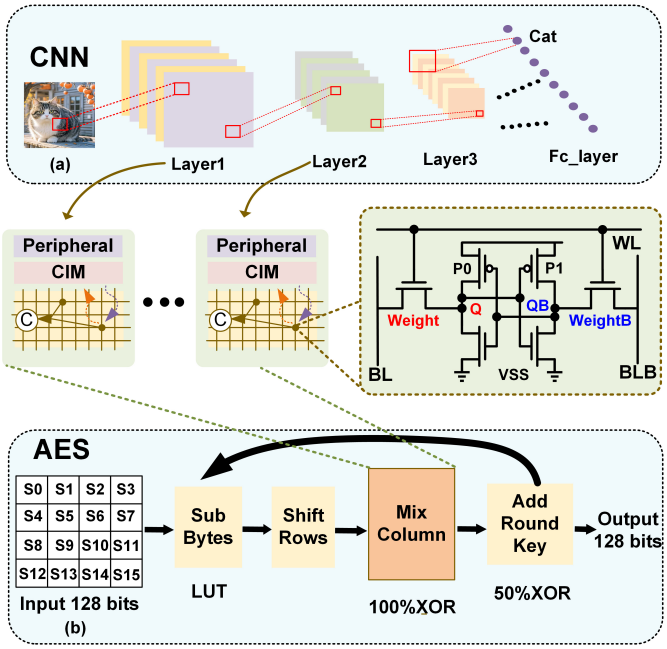


Fig. 7. Implementation of (a) CNN and (b) AES on Multiple SRAM Arrays

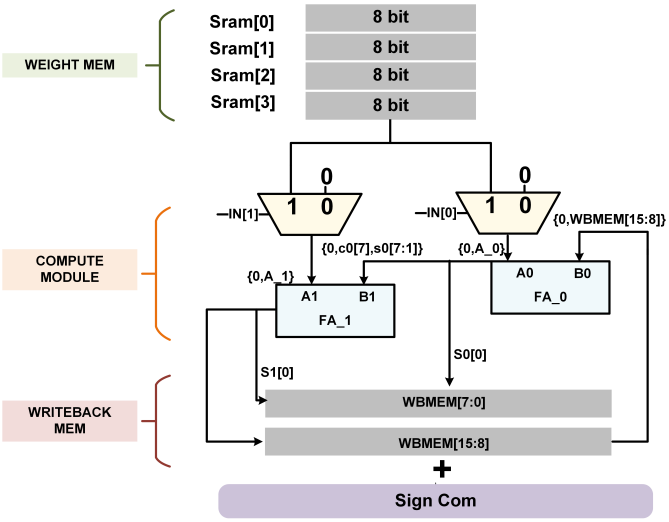


Fig. 8. 1-bit CIM Architecture Diagram

communication with the system bus.

B. Application of CIM in CNN and AES Algorithms

Edge AI chips focus on efficiently running neural network inference, with CNN as the main computational load. Among CNN operators, convolution accounts for over 50% of computation, making it the key target for acceleration. Convolution involves sliding-window local feature extraction, essentially performing matrix-vector multiplications [14], which suits the parallelism of CIM architectures—ideal for CIM-based acceleration.

Simultaneously, Advanced Encryption Standard (AES) is crucial for secure data transmission in edge AI, involving round key generation and iterative rounds with SubBytes, ShiftRows, MixColumns, and AddRoundKey operations. CIM architectures benefit AES implementation by embedding S-boxes in memory for in-situ substitution, leveraging memory parallelism for ShiftRows and MixColumns, and performing AddRoundKey XOR operations directly within memory—reducing data movement and power consumption [15], [16].

As shown in Figure 7, the proposed SRAM-based CIM convolution accelerator combines optimized hardware and dataflow scheduling to achieve energy-efficient computation. Integrating convolution acceleration and AES encryption in one CIM architecture enhances neural network inference performance while ensuring secure processing on edge AI chips [17].

III. Accelerator Design for AI

A. Convolution Operator Acceleration

This work proposes a CIM architecture utilizing 8-bit SRAM as WEIGHT MEM. As shown in Figure 8, each CIM BLOCK comprises a WEIGHT MEM, COMPUTE MODULE, and WRITEBACK MEM. Nine CIM BLOCKs, together with nine Sign Com units, form one CIM TOP to execute signed 3×3 convolutions. The complete CIM IP integrates 64 such CIM TOPs.

Each WEIGHT MEM includes four 8-bit SRAMs, maintaining a 4:1 ratio with the compute and writeback logic. This design allows efficient storage of 64 sets of 3×3 kernels using only 16×9 COMPUTE MODULEs, WRITEBACK MEMs, and Sign Com units, thereby reducing area.

Figure 9 shows the weight mapping scheme: one input FEATURE corresponds to four kernels (WEIGHT0–3), with their weights stored across Sram[0]–[3] in CIM BLOCK0–8. During computation, these weights are time-multiplexed and processed by the compute units to perform 3×3 convolutions.

B. Bus Interface Design and Data Flow Optimization of CIM Core

The interaction between the CIM core and the system bus, along with data flow scheduling, is crucial for the energy efficiency of edge AI chips. The CIM IP designed in this

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:04:38 UTC from IEEE Xplore. Restrictions apply.

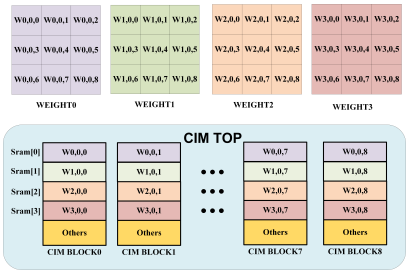


Fig. 9. Weight mapping in CIM TOP

paper includes 64 CIM TOP units, capable of storing 64 8-bit signed 3×3 convolution kernels. It connects to the bus via an AHB controller, with its configuration registers, input Cache (4×4×8bit), and output Cache forming an efficient interface architecture.

As shown in Figure 10, the input Cache is optimized for convolutional sliding windows, storing 4×4 feature subarrays and reusing overlapping data to reduce data movement. Combined with a "compute-while-update" pipeline mechanism, it shortens latency and reduces controller area [18]–[20].

Data flow follows the "storage-computation-transmission" collaboration principle. As illustrated in Figure 11(a), weights are written to SRAM, inputs are stored in Cache, and intermediate results after computation are temporarily stored in the output Cache. Figure 11(b) shows that when reaching the threshold, data is moved to QSPI FLASH by DMA. The CIM can switch to normal SRAM mode, and DMA reduces CPU intervention, improving bus bandwidth utilization.

This design reduces data movement energy consumption by 37% and computation latency by 29%, providing support for the high-energy-efficiency operation of edge AI chips.

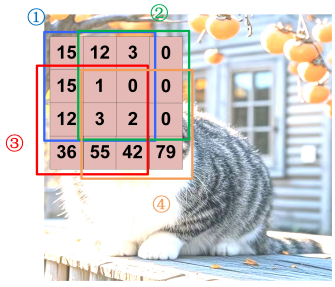


Fig. 10. sliding window convolution

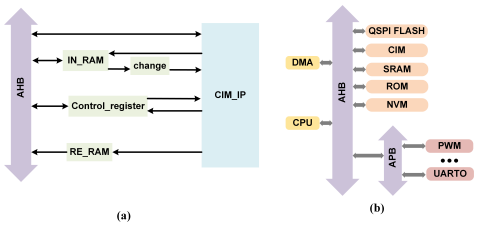


Fig. 11. (a) Connection relationship between CIM IP and bus (b) Data flow diagram of high-efficiency edge AI chip

IV. Conclusion

This paper combines the energy efficiency of ACIM for low-precision parallel operations with the high-precision advantages of DCIM to balance performance and efficiency. It also optimizes data flow scheduling (including sliding window caching and pipelined updates) to reduce data movement, which is a major source of energy consumption in edge devices. Experimental results show that it reduces data movement energy consumption by 37% and shortens latency by 29%, effectively supporting tasks such as CNN inference. Future work will focus on improving scalability for large neural networks and enhancing robustness against device variations, providing an efficient, secure, and high-performance development path for next-generation edge AI chips.

References

[1] P. -C. Wu et al., "A 22nm 832Kb Hybrid-Domain Floating-Point SRAM In-Memory-Compute Macro with 16.2-70.2TFLOPS/W for High-Accuracy AI-Edge Devices," 2023 IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, 2023, pp. 126-128, doi: 10.1109/ISSCC42615.2023.10067527.

[2] Z. Li, F. Liu, W. Yang, S. Peng and J. Zhou, "A Survey of Convolutional Neural Networks: Analysis, Applications, and Prospects," in IEEE Transactions on Neural Networks and Learning Systems, vol. 33, no. 12, pp. 6999-7019, Dec. 2022, doi: 10.1109/TNNLS.2021.3084827.

[3] H. Fujiwara et al., "A 5-nm 254-TOPS/W 221-TOPS/mm2 Fully-Digital Computing-in-Memory Macro Supporting Wide-Range Dynamic-Voltage-Frequency Scaling and Simultaneous MAC and Write Operations," 2022 IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, 2022, pp. 1-3, doi: 10.1109/ISSCC42614.2022.9731754.

[4] H. Mori et al., "A 4nm 6163-TOPS/W/b $\mathbf{4790-TOPS/mm^2/b}$ SRAM Based Digital-Computing-in-Memory Macro Supporting Bit-Width Flexibility and Simultaneous MAC and Weight Update," 2023 IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, 2023, pp. 132-134, doi: 10.1109/ISSCC42615.2023.10067555.

[5] S. -E. Hsieh et al., "7.6 A 70.85-86.27TOPS/W PVT-Insensitive 8b Word-Wise ACIM with Post-Processing Relaxation," 2023 IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, 2023, pp. 136-138, doi: 10.1109/ISSCC42615.2023.10067335.

[6] C. -F. Lee et al., "A 12nm 121-TOPS/W 41.6-TOPS/mm2 All Digital Full Precision SRAM-based Compute-in-Memory with Configurable Bit-width For AI Edge Applications," 2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits), Honolulu, HI, USA, 2022, pp. 24-25, doi: 10.1109/VLSITechnologyandCir46769.2022.9830438.

[7] W. Sun et al., "A Survey of Computing-in-Memory Processor: From Circuit to Application," in IEEE Open Journal of the Solid-State Circuits Society, vol. 4, pp. 25-42, 2024, doi: 10.1109/OJSSCS.2023.3328290.

[8] P. Chen et al., "A 22-nm Delta-Sigma Computing-In-Memory SRAM Macro With Near-Zero-Mean Outputs and LSB-First ADCs for Edge AI Processing," in IEEE Journal of Solid-State Circuits, vol. 60, no. 8, pp. 3020-3032, Aug. 2025, doi: 10.1109/JSSC.2025.3539736.

[9] P. Mannocci et al., "An SRAM-based reconfigurable analog in-memory computing circuit for solving linear algebra problems," 2023 International Electron Devices Meeting (IEDM), San Francisco, CA, USA, 2023, pp. 1-4, doi: 10.1109/IEDM45741.2023.10413724.

[10] V. Sze, Y. -H. Chen, T. -J. Yang and J. S. Emer, "Efficient Processing of Deep Neural Networks: A Tutorial and Survey," in Proceedings of the IEEE, vol. 105, no. 12, pp. 2295-2329, Dec. 2017, doi: 10.1109/JPROC.2017.2761740.

[11] A. Biswas and A. P. Chandrakasan, "Conv-RAM: An energy-efficient SRAM with embedded convolution computation for low-power CNN-based machine learning applications," 2018 IEEE International Solid-State Circuits Conference - (ISSCC), San Francisco, CA, USA, 2018, pp. 488-490, doi: 10.1109/ISSCC.2018.8310397.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:04:38 UTC from IEEE Xplore. Restrictions apply.

[12] X. Si et al., "A Twin-8T SRAM Computation-in-Memory Unit-Macro for Multibit CNN-Based AI Edge Processors," in IEEE Journal of Solid-State Circuits, vol. 55, no. 1, pp. 189-202, Jan. 2020, doi: 10.1109/JSSC.2019.2952773.

[13] Y. Zhan, W. -H. Yu, K. -F. Un, R. P. Martins and P. -I. Mak, "GSLP-CIM: A 28-nm Globally Systolic and Locally Parallel CNN/Transformer Accelerator With Scalable and Reconfigurable eDRAM Compute-in-Memory Macro for Flexible Dataflow," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 72, no. 4, pp. 1657-1667, April 2025, doi: 10.1109/TCSI.2024.3497187.

[14] F. Tu et al., "TranCIM: Full-Digital Bitline-Transpose CIM-based Sparse Transformer Accelerator With Pipeline/Parallel Reconfigurable Modes," in IEEE Journal of Solid-State Circuits, vol. 58, no. 6, pp. 1798-1809, June 2023, doi: 10.1109/JSSC.2022.3213542.

[15] J. Yue et al., "A 28nm 16.9-300TOPS/W Computing-in-Memory Processor Supporting Floating-Point NN Inference/Training with Intensive-CIM Sparse-Digital Architecture," 2023 IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, 2023, pp. 1-3, doi: 10.1109/ISSCC42615.2023.10067779.

[16] Z. Lin, Y. Liu, Y. Wang, Y. Zhao, C. Peng and X. Wu, "SRAM-Based Digital CIM Macro for Linear Interpolation and MAC," 2024 IEEE International Symposium on Circuits and Systems (ISCAS), Singapore, Singapore, 2024, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558525.

[17] X. Si et al., "A Local Computing Cell and 6T SRAM-Based Computing-in-Memory Macro With 8-b MAC Operation for Edge AI Chips," in IEEE Journal of Solid-State Circuits, vol. 56, no. 9, pp. 2817-2831, Sept. 2021, doi: 10.1109/JSSC.2021.3073254.

[18] M. Wu et al., "S2D-CIM: A 22nm 128Kb Systolic Digital Compute-in-Memory Macro with Domino Data Path for Flexible Vector Operation and 2-D Weight Update in Edge AI Applications," 2024 IEEE Custom Integrated Circuits Conference (CICC), Denver, CO, USA, 2024, pp. 1-2, doi: 10.1109/CICC60959.2024.10529046.

[19] H. You, W. Li, D. Shang, Y. Zhou and S. Qiao, "A 1–8b Reconfigurable Digital SRAM Compute-in-Memory Macro for Processing Neural Networks," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 71, no. 4, pp. 1602-1614, April 2024, doi: 10.1109/TCSI.2024.3355944.

[20] Y. Liu et al., "TSCIM: A 28nm Transposed Stochastic CIM Macro for On-Chip Training and Inference," 2025 IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025, pp. 1-5, doi: 10.1109/ISCAS56072.2025.11044117.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:04:38 UTC from IEEE Xplore. Restrictions apply.